SQL 4

ISIN 325 - Database Security

Will Schriner

1) Define and populate Rate and Customer tables in two separate databases.

RateID	BegAge	EndAge	Rate	Gender
1	0	4	12.2000	F
2	5	9	3.4100	F
3	10	14	3.9200	F
4	15	19	4.2200	F
5	20	24	4.3800	F
6	25	29	5.6600	F
7	30	34	6.3900	F
8	35	39	8.5200	F
9	40	44	10.1700	F
10	45	49	13.2200	F
11	50	54	21.2200	F
12	55	59	32.2300	F
13	60	64	40.1200	F
14	65	69	63.2200	F
15	70	74	113.4300	F
16	75	79	222.5300	F
17	80	84	557.4200	F
18	85	89	932.1200	F
19	0	4	13.0000	M
20	5	9	4.0000	M
21	10	14	4.5000	M
22	15	19	4.5700	M
23	20	24	4.7000	M
24	25	29	6.0000	M
25	30	34	7.0000	M
26	35	39	9.0000	M
27	40	44	10.7000	M
28	45	50	15.6700	M

2-IN-1\SQLE...bo.Customer | 2-IN-1\SQLEX...te - dbo.Rate | cust age view ...-1\wills (84))

CustID	FName	LName	DOB	Rider	Gender
1	James	Harrington	1985-03-14	Y	M
2	Sofia	Delacroix	1992-07-22	NULL	F
3	Marcus	Tillman	1978-11-05	Y	M
4	Priya	Nair	2001-01-30	NULL	F
5	Derek	Okafor	1967-09-18	Y	M
6	Hannah	Wexler	1995-04-03	NULL	F
7	Carlos	Mendoza	1988-12-27	NULL	M
8	Leah	Fontaine	2003-06-11	NULL	F
9	Raymond	Kowalski	1955-08-09	Y	M
10	Tanya	Osei	1973-02-16	Y	F
NULL	NULL	NULL	NULL	NULL	NULL

2) Include BeginAge and EndAge columns for the Rate table. The age ranges go in 5 year increments up to age 99. Separate records for male and female.

RateID	BegAge	EndAge	Rate	Gender
1	0	4	12.2000	F
2	5	9	3.4100	F
3	10	14	3.9200	F
4	15	19	4.2200	F
5	20	24	4.3800	F
6	25	29	5.6600	F
7	30	34	6.3900	F
8	35	39	8.5200	F
9	40	44	10.1700	F
10	45	49	13.2200	F
11	50	54	21.2200	F
12	55	59	32.2300	F
13	60	64	40.1200	F
14	65	69	63.2200	F
15	70	74	113.4300	F
16	75	79	222.5300	F
17	80	84	557.4200	F
18	85	89	932.1200	F

RateID	BegAge	EndAge	Rate	Gender
19	0	4	13.0000	M
20	5	9	4.0000	M
21	10	14	4.5000	M
22	15	19	4.5700	M
23	20	24	4.7000	M
24	25	29	6.0000	M
25	30	34	7.0000	M
26	35	39	9.0000	M
27	40	44	10.7000	M
28	45	50	15.6700	M
29	55	59	23.4500	M
30	60	64	45.0000	M
31	65	69	67.6700	M
32	70	74	125.3400	M
33	75	79	250.6500	M
34	80	84	605.0000	M
35	85	89	1100.2300	M

3) Create SQL routines that perform the following:

- assign people an insurance rate based on birthdate (age) and gender.

- Alter Customer table to include Rider flag. Compute and display rider cost - rider will increase rate by 1.5.

- Report of all customers sorted by age range, LName, FName. Displays the Rate/Rate x Rider

```sql
--creates a view
create view vw_GetCustInfo2 as

--calculation of age
Select CustID, FName, LName, DOB, Gender, Rider,
    datediff(year, DOB, GetDate())
    - case
        when dateadd(Year, DateDiff(Year, DOB, GetDate()), DOB) > getDate()
        then 1
        else 0
    end as age
from Customer
```

```sql
select c.custid, c.fname, c.lname, c.dob, C.Gender, c.rider, c.age, r.rate,
    case
        --Identifies when the rate should be multiplied
        when rider = 'y' then r.rate * 1.5
        else r.rate
    end as NewRate
--uses view created before
from vw_GetCustInfo2 C, Rate.dbo.Rate R
where c.age >= r.begage and c.age<= r.endage
and C.Gender = R.Gender
```

95 % No issues found

Results Messages

	custid	fname	lname	dob	Gender	rider	age	rate	NewRate
1	1	James	Harrington	1985-03-14	M	Y	41	10.70	16.05000
2	2	Sofia	Delacroix	1992-07-22	F	NULL	33	6.39	6.39000
3	3	Marcus	Tillman	1978-11-05	M	Y	47	15.67	23.50500
4	4	Priya	Nair	2001-01-30	F	NULL	25	5.66	5.66000
5	5	Derek	Okafor	1967-09-18	M	Y	58	23.45	35.17500
6	6	Hannah	Wexler	1995-04-03	F	NULL	31	6.39	6.39000
7	7	Carlos	Mendoza	1988-12-27	M	NULL	37	9.00	9.00000
8	8	Leah	Fontaine	2003-06-11	F	NULL	22	4.38	4.38000
9	9	Raymond	Kowalski	1955-08-09	M	Y	70	125.34	188.01000
10	10	Tanya	Osei	1973-02-16	F	Y	53	21.22	31.83000